

Which fifteen-minutes neighborhoods are dead-ends? An analysis of the network attributes of fifteen-minute pedsheds

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Abstract

Fifteen-minutes neighborhoods, a form of normative chronourbanism based on cumulative opportunities, has aroused interest as a way to reduce the need for motorized travel, and increase the livability, convenience, and health of the public. At the core of this concept is a pedshed, an area defined by the walkable isochrone of the eponymous fifteen minutes. As the idea of fifteen-minutes neighborhoods develops traction in policy and planning circles, it seems timely to revisit the way street network design can support—or obstruct—the stated goal of preserving or creating walkable neighborhoods with essential amenities. In this paper we examine a sample ($n = 834$) of fifteen-minutes pedsheds in Hamilton, a medium-sized city in Canada, and how their sizes relate to the attributes of the transportation network. The analysis reveals that network design in suburban Hamilton conspires against the creating of fifteen-minutes neighborhoods. Much of urban Hamilton, in contrast, already has the characteristics of fifteen-minutes neighborhoods. The research points to elements of network design that can help to discriminate between candidate neighborhoods and dead-ends, and that can provide parameters for the design of new developments.

Keywords: Fifteen-minutes neighborhoods, Pedshed, Walkability, Accessibility, Network analysis

1. Introduction

“Il faut oublier la traversée de Paris d’est en ouest en voiture”³. –Anne Hidalgo

With this declaration during her reelection campaign in 2020, Anne Hidalgo brought international attention to the concept of walkable fifteen-minutes neighborhoods (Alimi et al., 2020).

Knight et al. (2018) Liu et al. (2022) Pozoukidou and Chatziyiannaki (2021) Weng et al. (2019)

We address the following research questions:

1. Is there a difference in the transportation network topology between urban and suburban areas within 15-minute walking distances?
2. If such a difference exists, what is its effect on the 15-minute walksheds in terms of accessibility, where accessibility is understood as the provision of opportunities?

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¹This is the first author footnote.

²Another author footnote.

³“We must forget about crossing Paris from east to west by car”

2. Data

Relevant data to develop this paper comprises to use 15-min pedestrian sheds (or walksheds) classified into urban and suburban types in Hamilton.

15-minute walksheds in the city of Hamilton were obtained by extracting relevant structural attributes of the street network within 15-minute pedestrian sheds. These sheds were calculated for each shed as polygon representations of origin-destination walking time matrices, with a maximum travel time of 15 minutes. Walkshed calculations were generated using the `travel_time_matrix` function for walking from the `{r5r}` R package. This process involved defining the street network, as well as the origins and destinations of walking trips. The street network data for these calculations was sourced from OpenStreetMap OpenStreetMap contributors (2023).

The origins we established are proxies for the centroids of neighborhoods in the urban and peri-urban areas of the city of Hamilton. Specifically, these are the centroids of Hamilton’s Dissemination Areas (DA). A DA is composed of one or more neighboring dissemination blocks and represents the smallest standard geographic area for which all census data are disseminated in Canada. Data on urban and sub-urban boundaries, DAs, and population data in 2016 were retrieved from Ham.

Destinations were established as the centroids of geohashed hexagons within the region of interest, where Hamilton’s amenities are located, assuming that presence of amenities is a proxy to provision of opportunities. These destinations were chosen to avoid calculating travel times to each individual amenity. Instead, we used the geohashed polygons as destinations, identifying the amenities within each polygon and assuming that the centroid of each hexagon represents the location of the amenities within it. Specifically, we retrieved geohashing data for Hamilton using the H3 system Kuethe and generated child hexagons at resolution 13, with an average area of 43.87 m^2 . Amenities location and their typology were downloaded from OpenStreetMap contributors (2023).

3. Methods

The proposed method involves studying 15-minute walking neighborhoods in two stages. In the first stage, we assess the accessibility of Hamilton’s walksheds from a normative perspective, focusing on the availability of opportunities. In the second stage, we analyze the topology of urban and suburban walksheds based on various network structure attributes. Finally, we examine whether there are differences in transportation network topology between urban and suburban areas, and analyze how these differences affect access to opportunities within walksheds.

3.1. Calculation of accessibility levels

The indicator we use assesses levels of accessibility from a normative perspective, focusing on reachable opportunities as estimates of the potential for spatial interactions Hansen (1959). The advantage of using this form of accessibility analysis in this study is that it has been shown to be a valuable method for exploring the relationship between transportation infrastructure and urban structure Handy and Niemeier (1997).

In this study, we calculate cumulative accessibility scores by summing the number of amenities within urban and suburban walksheds, categorized into the four types provided by the data source: financial, sustenance, healthcare, and library. Additionally, we extract population data of each walkshed. Summary statistics of population are also included in the analysis.

Additionally, the accessibility assessment is completed by obtaining summary statistics of travel times in urban and suburban walksheds. To study the impact of urban and suburban walksheds on walking decisions, minimum and mean travel times to destinations are introduced into time-decay functions for walking decisions, as developed by Milward (2013). These are active travel impedance functions based on data from Canada.

3.2. Urban and suburban walkshed modelling

3.2.1. Network attributes by walkshed

In this work we characterize urban and suburban walksheds based on their network structure. As a way to capture and describe the structure of networks we calculate a dplyr::selection of 12 network structural attributes using the network analysis package {igraph} (Csárdi et al., 2024). The set of attributes that we consider for this work is described Table 1.

Table 1: Network attributes in the analysis

attribute	description
Number of vertices	The total number of vertices in a graph.
Number of edges	Total number of edges in a graph.
Diameter	The longest shortest path in a graph, akin to the longest segment of a line that cuts a circle.
Radius	Minimum of all the maximum distances between each vertex and all other vertices.
Global efficiency	The average of the inverse distances between all pairs of vertices.
Edge density	The ratio of the number of edges in a graph to the maximum number of possible edges (i.e., the number of edges of a point-to-point network).
Mean distance	Mean length of all shortest paths from each vertex.
Girth	The length of the shortest cycle, where a cycle is a trail where only the starting and ending point are a repeated vertex.
Minimum cut	The smallest number of edges that, when removed, separate a graph into two or more components.
Edge connectivity	Minimum number of nodes that, when removed, disconnect two nodes.
Transitivity	Probability that a vertex neighbors are neighbors between them.
Motifs (v)	Isomorphic subgraphs when the number of nodes is (v). A graph with few motifs will have many unique elements; a graph with a large number of motifs will have a small number of patterns that are repeated. In this analysis we calculate motifs of size v=3 and v=4.

3.2.2. Modelling

To model urban and suburban walksheds, we employ a classification tree approach using the set topology attributes calculated in the previous stage. Classification trees are machine learning techniques used for both regression and classification, widely appreciated in various disciplines for their ability to represent knowledge through decision rules, making them easy to interpret and analyze Loh (2011) James et al. (2013) Paez et al. (2019). Specifically, we utilize evolutionary trees, a variant of decision trees that are less constrained by the limitations of greedy top-down induction algorithms Cha and Tappert (2009) Kretowski and Grześ (2007) Grubinger et al. (2014). Since evolutionary trees use a genetic algorithm that introduces randomness in the initial conditions, we run the model from the R package evtree Grubinger et al. (2014) with 20 random seeds to identify the network variables that consistently characterize urban and suburban walksheds.

Finally we study effects in accessibility

3.3. Walkshed size in urban and suburban areas

4. Results and discussion

4.1. Classification of walksheds based on network attributes

Evolutionary trees: discuss training procedure. We used a training sample and a test sample. We trained n=30 trees per batch, and saved the results for the batch with the best tree (5 batches). The best tree was dplyr::selected based on the evaluation function (write equation and explain).

The results of training an evolutionary classification tree are shown in Fig. 1. This model addresses our first research question (are networks different between urban and suburban regions?). The answer to this question is unambiguously affirmative. As seen in the figure, the tree has four terminal nodes (or leaves). The purity of the leaves (meaning the mix of classes in a leaf) is very high, since each leaf is composed largely of a single class (77.4% and 80.2% Suburban in Leaves 1 and 3 respectively, and 88.8% and 87.9% Urban in Leaves 2 and 4 respectively). Thus, there are two kinds each of urban and suburban walksheds, depending on the attributes of the network that generated them.

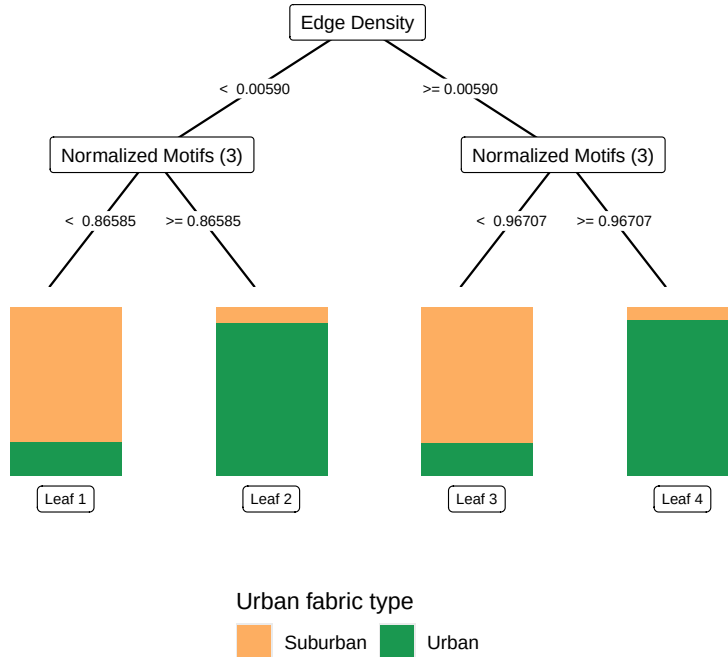


Figure 1: Evolutionary classification tree. The dependent variable is the fabric type: Suburban (orange), Urban (green).

Of the attributes in Table 1, only two are picked up by the classification algorithm: edge density and motifs of size 3. The first attribute that helps to discriminate between urban and suburban walksheds is edge density. The cutoff value of edge density is 0.00590. Recall that edge density is the number of edges in the graph relative to the maximum possible number of edges—that is, compared to a fully connected graph. This means that the cutoff is about 0.59% of the number of edges in a fully connected graph with the same number of vertices. The range of values of this variable in our walksheds is 0.117% and 40% (the mean is 1.246%). But there are marked differences in the value of this attribute by Leaf in the classification tree, as seen in Table 2

Table 2: Descriptive statistics of edge density by Leaf in classification tree.

leaf	leaf description	mean	s.d.	min	first quartile	median	third quartile	max
Leaf 1	Suburban (Type I: low edge density)	0.00507	0.00056	0.00323	0.00469	0.00517	0.00550	0.00589
Leaf 2	Urban (Type I: low edge density)	0.00354	0.00128	0.00117	0.00254	0.00362	0.00450	0.00590
Leaf 3	Suburban (Type II: high edge density)	0.01654	0.02482	0.00591	0.00757	0.01018	0.01631	0.40000
Leaf 4	Urban (Type II: high edge density)	0.01092	0.00869	0.00591	0.00633	0.00710	0.01057	0.05326

Table 3 shows the values of the motifs of size 3 by Leaf.

Table 3: Descriptive statistics of motifs of size 3 by Leaf in classification tree.

leaf	leaf description	mean	s.d.	min	first quartile	median	third quartile	max
Leaf 1	Suburban (Type I: low edge density)	0.84626	0.01615	0.79574	0.84063	0.85155	0.85727	0.86562
Leaf 2	Urban (Type I: low edge density)	0.96914	0.04703	0.86585	0.93915	0.96858	1.00302	1.09044
Leaf 3	Suburban (Type II: high edge density)	0.84114	0.06536	0.33333	0.80603	0.84407	0.88309	0.96646
Leaf 4	Urban (Type II: high edge density)	1.00809	0.02705	0.96774	0.98674	1.00453	1.03274	1.05860

The differences between walksheds in various types of urban and suburban fabric are better appreciated by means of some examples. Fig. REF

Where are these found?

DISCUSS: there appear to be significant (?) differences between the sizes of the walksheds.

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4.2. Area of 15-min walksheds

Walksheds in suburban areas tend to be smaller, walksheds in urban areas larger. Are there differences between walksheds of Type I and of Type II? Test differences of means.

Table 4: Descriptive statistics of walkshed area by Leaf class.

leaf	leaf description	mean	s.d.	min	first quartile	median	third quartile	max
Leaf 1	Suburban (Type I: low edge density)	0.50306	0.07392	0.30974	0.45472	0.51397	0.55072	0.67990
Leaf 2	Urban (Type I: low edge density)	0.57102	0.08205	0.13681	0.52480	0.58866	0.62817	0.73116

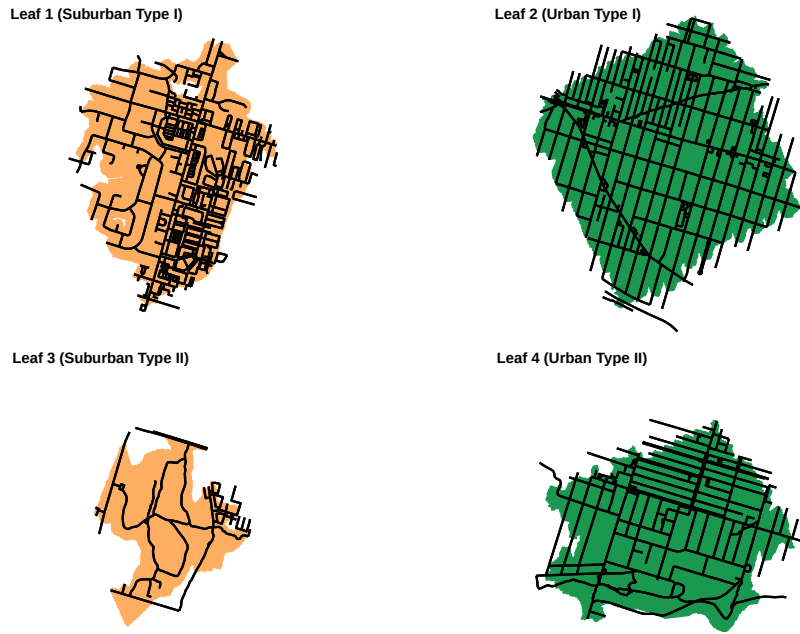


Figure 2: Examples of walksheds and their networks by Leaf in the classifications.

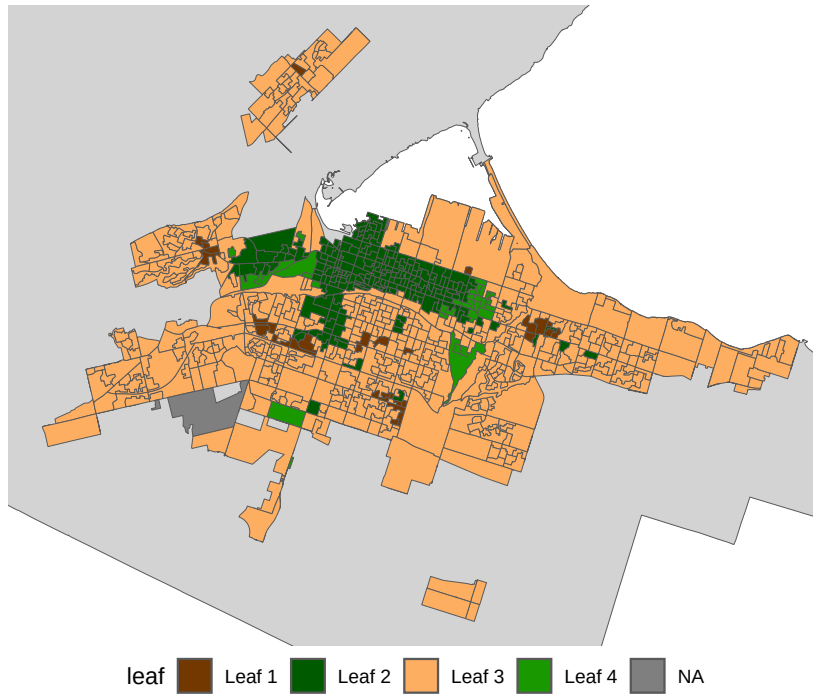


Figure 3: Distribution of Leaf classes in urban and suburban Hamilton (gray area is rural Hamilton).

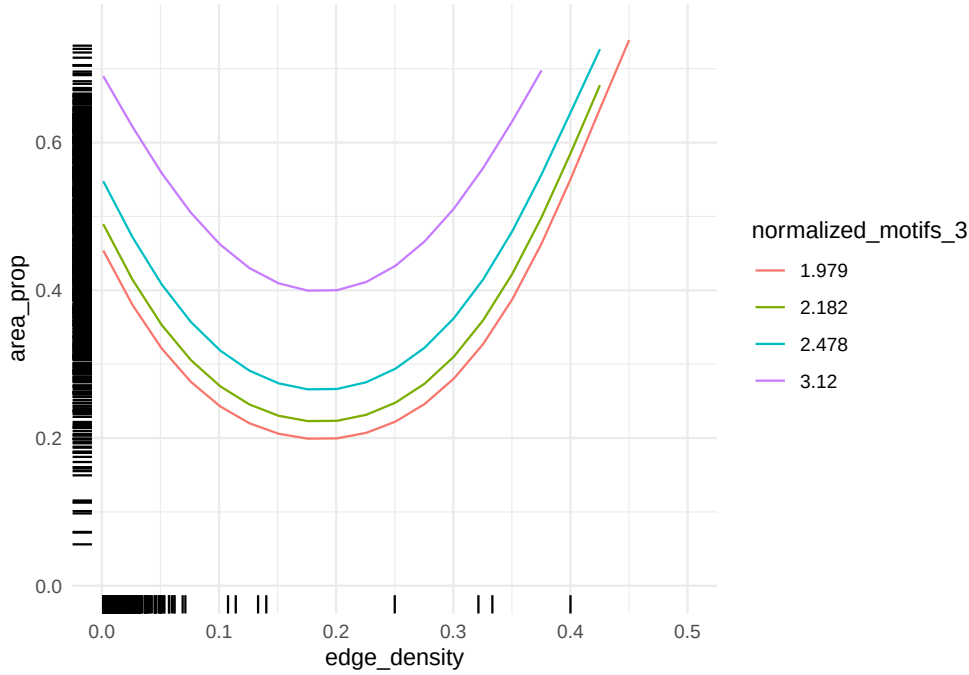


Figure 4: Estimated proportion of maximum area as a function of edge density and normalized number of motifs of size 3.

```
## (Intercept)          -2.104e+00  1.468e-01 -14.329 < 2e-16 ***
## edge_density         -1.313e+01  1.332e+00  -9.859 < 2e-16 ***
## sq_edge_density      3.514e+01  3.723e+00   9.437 < 2e-16 ***
## sq_normalized_motifs_3 1.691e-01  7.405e-03  22.838 < 2e-16 ***
## transitivity         1.095e+00  4.501e-01   2.433  0.015 *
## global_efficiency    -1.412e+02  2.350e+01  -6.010 1.85e-09 ***
## mean_distance        1.870e-03  1.288e-04  14.512 < 2e-16 ***
##
## Phi coefficients (precision model with identity link):
##      Estimate Std. Error z value Pr(>|z|)
## (phi)   58.120      2.825   20.57  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Type of estimator: ML (maximum likelihood)
## Log-likelihood: 1120 on 8 Df
## Pseudo R-squared: 0.7479
## Number of iterations: 137 (BFGS) + 14 (Fisher scoring)
```

4.3. 15-min accessibility analysis

Another feature of suburban environments is that they tend to be lower density. Proceed to accessibility calculations.

Words go here.

4.4. Implications for walking behavior

Use Millward et al. (2013)'s impedance function(s) to discuss the implications of characteristics of walk-sheds for walking behavior, concretely that there is a difference in the mean time to reach destinations within

different types of walksheds.

5. Conclusions

Words go here.

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